HipMRI 3D Prostate Segmentation Feasibility Review

COMP3710 Hard difficulty 3D Improved U Net with residual units and deep supervision

Dataset	Difficulty	Target
HipMRI 3D NIfTI volumes	Hard	Prostate Dice at least 0.70
Baseline	Proposed	Split
Plain 3D U Net	3D Improved U Net	Patient disjoint; seed 42

User need scope and decision requested

A radiotherapy contouring specialist needs a reviewable 3D draft of prostate and organ-at-risk boundaries. The engineering question is whether volumetric context, residual units and deep supervision reduce apex, base, cross-slice discontinuity and bladder or rectum boundary errors enough to justify higher GPU memory and runtime. I request confirmation that this Hard HipMRI 3D scope and resource plan are appropriate before full training.

Method and acceptance criteria

The comparison uses a plain 3D U-Net baseline and a 3D Improved U-Net with volumetric residual units and decoder deep supervision. Both use one saved patient-level train, validation and test manifest. Acceptance evidence is: zero patient overlap; aligned NIfTI image-mask volumes; held-out prostate Dice at least 0.70; per-class Dice/IoU, peak VRAM and per-patient inference time; and 3-5 real failure cases with false-positive and false-negative overlays.

Preliminary feasibility evidence

Local synthetic-NIfTI tests currently verify recursive image-mask pairing, missing-pair rejection, 3D shape and affine validation, deterministic patient-disjoint splitting, finite 3D forward/backward passes, residual deep-supervision outputs, validation-only checkpoint selection and held-out metric fields. This establishes the software path only. HipMRI data audit, GPU smoke measurement and final Dice values are not yet claimed.

Risks budget and fallback

Main risks are 3D volume memory demand, small foreground regions, NIfTI orientation errors, slow I/O and failed cluster jobs. The next experiment is a short Rangpur GPU smoke run on audited volumes to set patch geometry, batch size and mixed-precision settings, followed by matched Slurm runs. If the improved model does not improve real held-out boundary failures at acceptable cost, the plain 3D U-Net remains the fallback recommendation.

Questions for tutor confirmation

1. Confirm the Hard HipMRI 3D Improved U-Net scope and patient-level evaluation protocol. 2. Confirm whether the prostate Dice >= 0.70 target is assessed with background reported separately. 3. Confirm the expected GPU budget, patch strategy and midpoint evidence required before full training.